



EC-201: Engineering Project Management

Concepts of Project Management

Broad Contents

- Project Management
- *Efficiency* and *effectiveness* in projects
- The project management system
- Project manager

3.1 What is Project Management? (1/4)

- Project Management is the discipline of organizing and managing resources in such a way that these resources deliver all the work required to complete a project within defined scope, time, and cost constraints. It is important to note here that a project is a temporary and one-time endeavour undertaken to create a unique product or service that brings about beneficial change or added value.
- This property of being a temporary and one-time undertaking contrasts with processes, or operations, which are permanent or semi-permanent ongoing functional work to create the same product or service over and over again.

3.1 What is Project Management? (2/4)

- The first challenge of project management is ensuring that a project is delivered within the defined constraints. The second, more ambitious, challenge is the optimized allocation and integration of the inputs needed to meet those predefined objectives. The project, therefore, is a carefully selected set of activities chosen to use resources (money, people, materials, energy, space, provisions, communication, quality, risks, etc.) in order to meet the objectives established by the organization.
- Management in any project is concerned with productivity. This refers to efficiency and effectiveness. These can be explained as follows:
 - **Efficiency:** In order to be efficient, management is concerned with minimizing resource costs. Efficiency is “doing things right”.
 - **Effectiveness:** In order to be effective, management is concerned with getting activities completed. Effectiveness is “doing right things”.

3.1 What is Project Management? (3/4)

- Thus, efficiency is concerned with means and effectiveness with ends. They are interrelated. It is easier to be effective if one ignores efficiency. For example, some organizations are reasonably effective, but are extremely inefficient. They get their jobs done, but at a very high cost.
- For the management of any project, it is important not only to get the activities completed(effectiveness), but also to do so as efficiently as possible. Can organizations be efficient and yet not effective? Yes, by doing wrong things well.
- The following figure (figure 3.1 next slide) shows management seeking efficiency and effectiveness.

3.1 What is Project Management? (4/4)

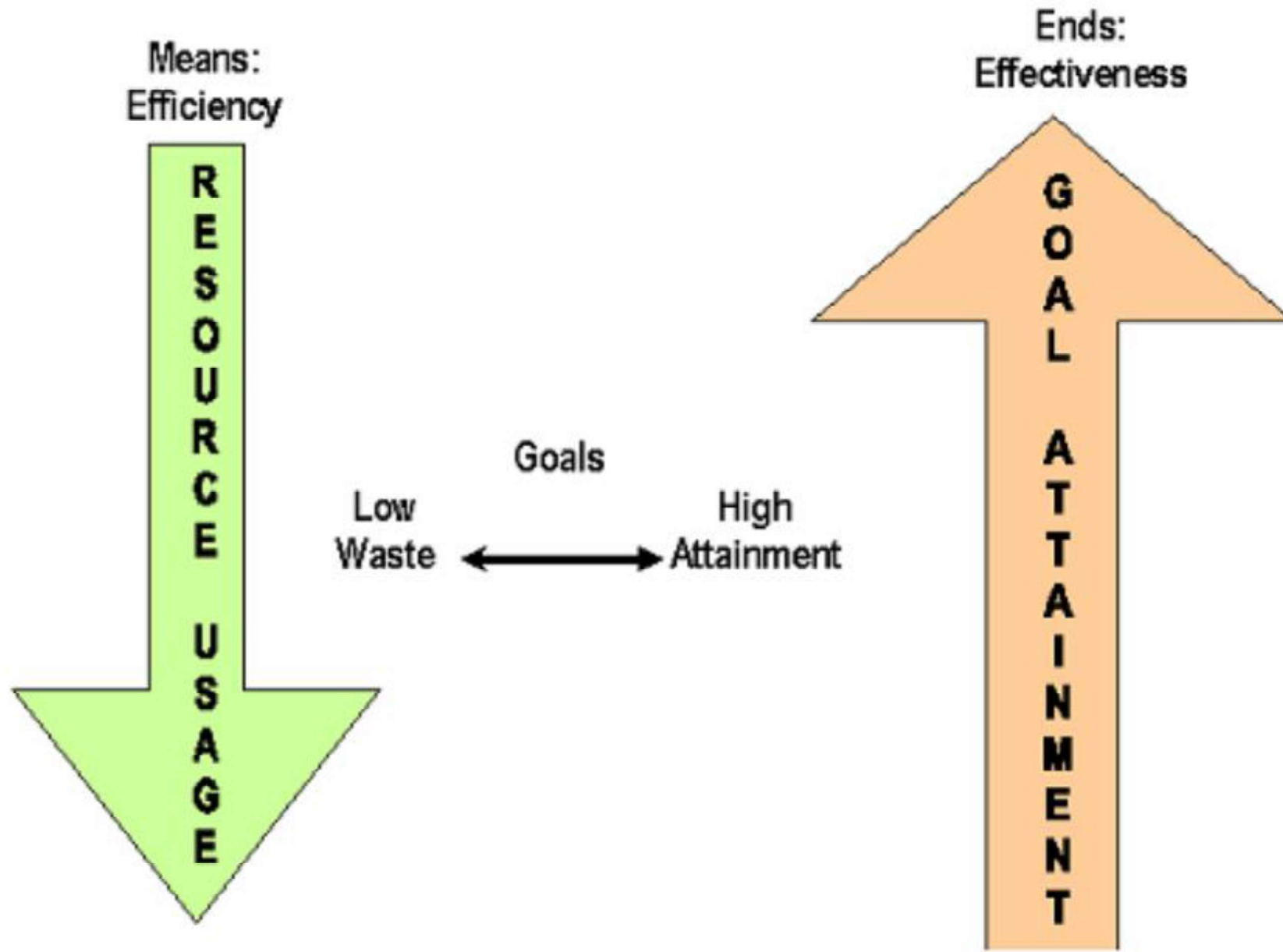


Figure 3.1: Efficiency and Effectiveness

3.2 The Project Management System (1 / 2)

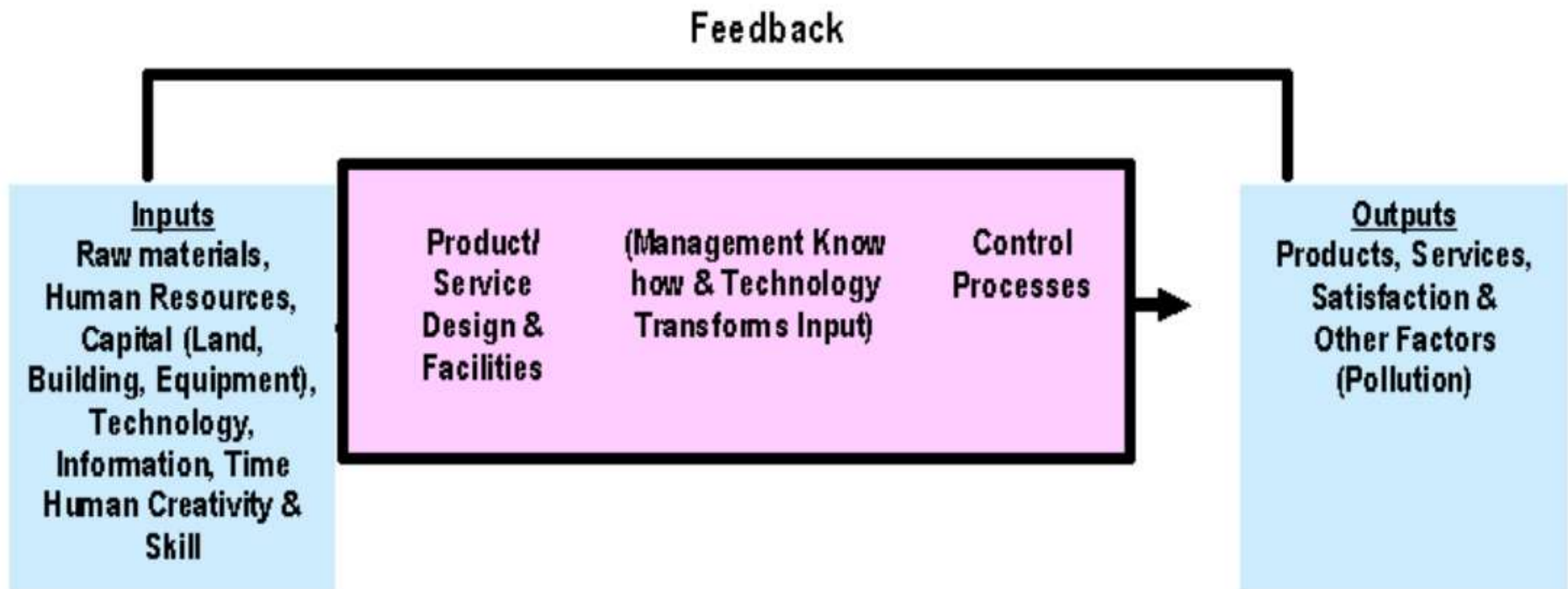


Figure 3.2: Project Management System

3.2 The Project Management System (2/2)

- Because of the interrelatedness of these driving forces, some people contend that the only true driving force is survival. This is illustrated in Figure 3.3 below. When the company recognizes that survival of the firm is at stake, the implementation of project management becomes easier. The speed by which companies reach some degree of maturity in project management is most often based upon how important they perceive the driving forces to be.

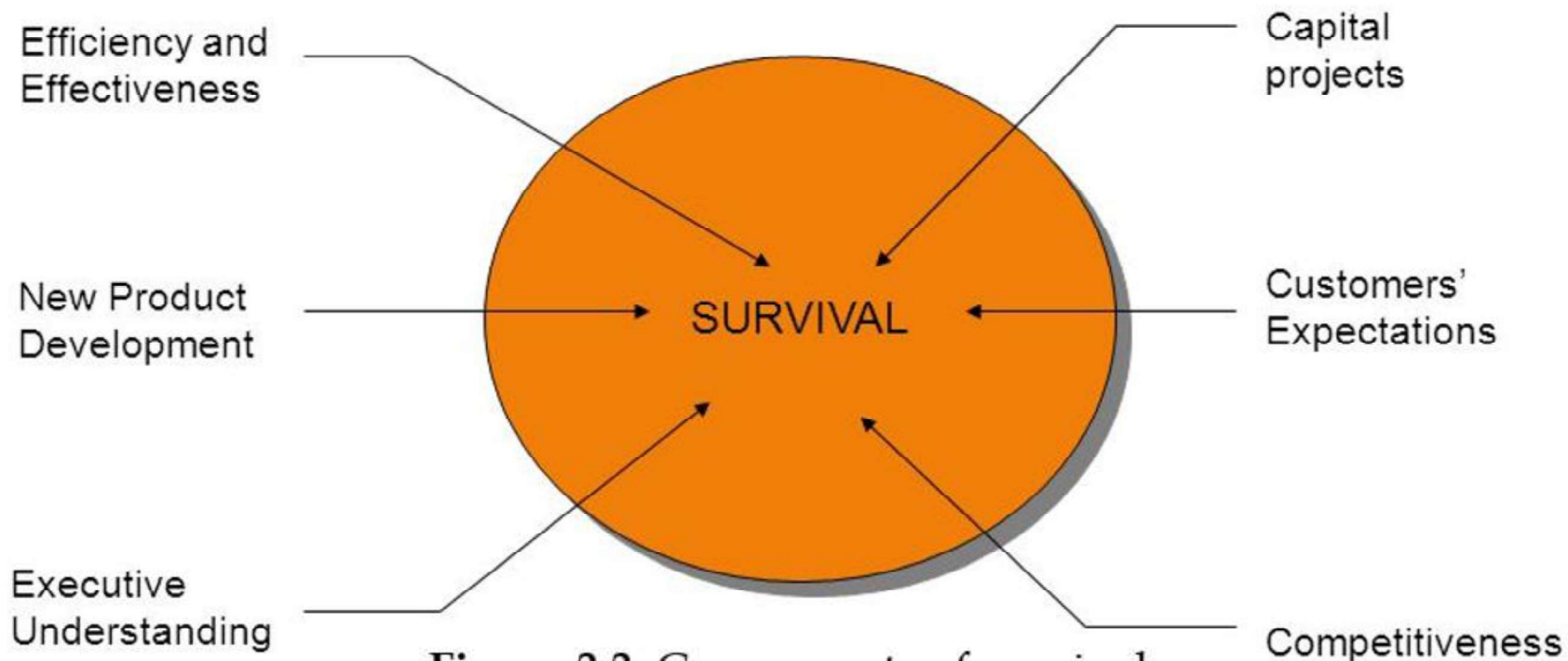


Figure 3.3: Components of survival

3.3 Who is a Project Manager? (1/5)

- A project manager is a professional in the field of project management. He has the responsibility of the planning and execution of any project. A project manager's central duty is to ensure the success of a project by minimizing risk throughout the lifetime of the project. This is done through a variety of methods, both formal and informal. A project manager usually has to ask penetrating questions, detect unstated assumptions, and resolve interpersonal conflicts, as well as use more systematic management skills.
- In whatever field, a successful project manager must be able to visualize the entire project from start to finish and should have the ability to ensure that this vision is realized.

3.3 Who is a Project Manager? (2/5)

a. *Types of Project Managers:* Project managers cannot perform their tasks well unless they have understanding of and are responsive to many elements of the external environment, including; economic, technological social, political and ethical factors that effect their areas of operations. The various types of project managers are follows:

- *Line managers* are responsible for activities making direct contributions to production of organization's basic goods or services.
- *Staff managers* use special technical expertise to advise and support the efforts of line workers.
- *Functional managers* are responsible for only one area of activity, i.e. finance, marketing, production, personnel, accounting, or sales.
- *General Managers* are responsible for complex organizational unit that include many areas of functional activity.
- An *administrator* is someone who administers work in any kind of organization.

3.3 Who is a Project Manager? (3/5)

b. *Activities of Project Managers:* Following are the four major activities that are undertaken by the project managers:

- 1. *Traditional management:*** This includes decision making, planning, and controlling.
- 2. *Communication:*** This refers to exchanging routine information and processing paperwork.
- 3. *Human Resource Management (HRM):*** It involves motivating, disciplining, managing conflict, staffing, and training.
- 4. *Networking:*** It includes socializing, and interacting with outsiders.

An average manager spends:

- 32% of time in traditional management activities
- 29% in communicating
- 20% in HRM activities
- 19% in networking

3.3 Who is a Project Manager? (4/5)

c. *Success for Project Managers:* There are three general preconditions for achieving lasting success as Project Manager. These include:

- *Ability (A)*
- *Motivation to manage (M)*
- *Opportunity (O)*

Together, they constitute the basic formula for managerial success (S):

$$S = A \times M \times O$$

3.3 Who is a Project Manager? (5/5)

d. Ten Facts of Project Managerial Life:

- i) Project Managers work long hours. Number of hours worked tends to increase as one climbs the managerial ladder.
- ii) Project Managers are busy. Typical manager's day is made up of hundreds of brief incidents or episodes. Activity rates tend to decrease as rank increases.
- iii) Project Manager's work is fragmented. They have little time to devote to any single activity. Interruptions and discontinuity are the rule. Project Manager's job is varied. They engage in variety of activities
- iv) (paperwork, phone calls, scheduled and unscheduled meetings, and inspection tours/visits). They interact with variety of people, and deal with variety of content areas.
- v) They spend most of their time pursuing activities within their own organizations. As managerial rank increases, they spend proportionately more time outside their work areas and organizations.
- vi) Project Manager's work is primarily oral. At all levels, they spend most of the time communicating verbally by personal contacts/ telephone etc.
- vii) Project Managers use a lot of contacts. Consistent with their high level of verbal communication, managers continually exchange information with superiors, peers, subordinates, and outsiders on ongoing basis.
- viii) Project Managers are not reflective planners. Typical manager is too busy to find uninterrupted blocks of time for reflective planning.
- ix) Information is the basic ingredient of Project Manager's work. Managers spend most of their time obtaining, interpreting, and giving information.
- x) Project Managers do not know how they spend their time. Managers consistently overestimate the time they spend on production, reading and writing, phone calls, thinking, and calculating and consistently underestimate time spent on meetings as well as on informal discussions.

3.3 Who is a Project Manager? (5/5)

e. *Managerial Skills*

A skill is an ability or proficiency in performing a particular task. Skills reflect the ability to translate actions into results. They are of the following types:

- ***Technical Skill*** is the knowledge of and proficiency in activities involving methods, processes, and procedures.
- ***Human Skill*** is the ability to work with people; cooperative effort; it is teamwork; feel secure and free to express their opinions.
- ***Conceptual Skill*** is the ability to see “big picture” in order to recognize significant elements in a situation, and to understand relationships among elements.
- ***Design Skill*** is the ability to solve problems in ways that will benefit enterprise.

3.4 Tomorrow's Management Today

- Average company will be smaller, employing fewer people.
- Traditional organizational structures will become more team-based and without boundaries.
- Employees will be empowered to make decisions.
- Work will be organized around teams and processes.
- Stress will be on vision and values.
- Managers will be change agents.
- Leadership will be more important.

References:

† Harold Kerzner, *“Project Management, A System Approach to Planning, Scheduling & Controlling”*, 12th Edition, Wiley.

† James. P. Lewis *“Project Planning, Scheduling, and Control”*, 3rd Edition, McGraw-Hill Companies.

† Timothy J. Kloppenborg, Vittal Anantatmula, Kathryn N. Wells, *“Contemporary Project Management”* 4th Edition, Cengage Learning.

